



InteriorVision: Classification and Generation of Interior Design Images

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Abstract

This project presents a dual-track deep learning framework for interior design analysis and generation. The system integrates two complementary models trained on a curated dataset of 4,000 images: a discriminative track employing an enhanced multi-task ResNet50 for image classification, and a generative track utilizing Stable Diffusion with Low-Rank Adaptation (LoRA) for text-to-image synthesis.

To improve classification, the baseline CNN was optimized via full backbone fine-tuning, dropout regularization, batch normalization, and label smoothing. The enhanced CNN achieved 74% room-type accuracy and 60% style accuracy, significantly outperforming the baseline. For the generative task, the LoRA-augmented Stable Diffusion model successfully synthesized domain-specific interior concepts from text prompts. While qualitative results demonstrated high stylistic consistency, quantitative evaluation via CLIP scores highlighted the strict dependencies of diffusion models on dataset scale and hardware constraints.

Overall, this framework demonstrates the efficacy of combining discriminative and generative deep learning pipelines for specialized design domains under restricted data and hardware regimes.

1 Introduction

1.1 Problem Statement:

Interior design images exhibit high variability in lighting, textures, spatial layout, and stylistic elements. Traditional computer vision techniques struggle to capture these abstract and multi-level patterns, which limits their ability to classify or generate realistic interior scenes. The challenge lies in learning both local visual cues (e.g., edges, materials, furniture shapes) and global semantic structure (e.g., room layout, style coherence).

Deep learning provides a scalable solution by automatically learning hierarchical representations that can generalize across diverse interior environments.

1.2 Why Deep Learning is Suitable

Deep learning models particularly Convolutional Neural Networks (CNNs) and Latent Diffusion Models are well-suited for this domain because they:

- Extract multi-level spatial features directly from raw images.
- Capture global context such as style, color harmony, and room composition.
- Learn complex mappings between text prompts and visual outputs.

- Adapt efficiently to new data distributions through fine-tuning.
- This makes deep learning ideal for both discriminative tasks (classification) and generative tasks (image synthesis).

1.3 Objectives

The main goal of this project is to build, analyze, and improve two different computational tracks to handle our image dataset. The specific objectives are divided into two main parts:

The Discriminative Track (Image Classification)

- Implement **ResNet50** as the baseline classification model.
- Analyze how bottleneck blocks and residual skip connections extract deep features without vanishing gradients.
- Evaluate classification performance using Global Average Pooling (GAP) and Softmax.

The Generative Track (Image Synthesis)

- Deploy **Stable Diffusion** as the baseline for text-to-image generation.
- Study how a Variational Autoencoder (VAE) compresses images into an efficient latent space.

Optimization & Enhancements

- Propose and test structural and algorithmic improvements for both baseline models.
- Enhance training speed (convergence), generalizability, and overall performance metrics.

2 Dataset

2.1 Dataset Description

This project uses the Interior Design Images & Metadata Dataset from Kaggle. The dataset contains interior design images grouped by room type and design style. It was chosen because it provides both images and structured labels, which makes it suitable for the classification and image generation parts of the project.

The dataset includes 4,139 images from four room categories: bathroom, bedroom, kitchen, and living room. It also includes five design styles: boho, industrial, minimalist, modern, and Scandinavian. Each image has metadata such as the image path, room type, style, and encoded label.

The dataset is already divided into training, validation, and testing sets, which helps us train and evaluate the models in an organized way.

In this project, the dataset is used for two main tasks. First, it is used to train CNN models to classify the room type and interior style. Second, the room type and style labels are used to create text prompts for the image generation model. These prompts help the Stable Diffusion model generate interior design images that match the given description.

Table 1: Summary of dataset splits before and after the data cleaning process

Split	Before Cleaning	After Cleaning
Training	2,648	2,556
Validation	663	639
Testing	828	805
Total	4,139	4,000

2.2 Data Preprocessing

Before training the models, several preprocessing steps were applied to prepare the dataset and make it suitable for both classification and image generation tasks.

2.2.1 Path Correction and Cleaning.

The image paths provided in the CSV files were checked and corrected to match the working directory used in Google Colab. Images with invalid or missing paths were removed when necessary to avoid loading errors during training.

2.2.2 Prompt Generation.

For the generative part of the project, text prompts were automatically created from the metadata by combining the design style and room type. For example, an image labeled as modern and kitchen was converted into the prompt: “modern kitchen interior”. These prompts were used to guide the Stable Diffusion model during image generation and LoRA fine-tuning.

2.2.3 Image Resizing and Format Conversion.

All images were converted to RGB format and resized to a fixed size before being passed to the

models. The classification model used resized images suitable for the CNN input, while the generative model used images prepared for Stable Diffusion training.

2.2.4 Normalization and Augmentation.

For the CNN classifier, image pixel values were normalized before training. In the improved model, data augmentation was applied to the training images to increase diversity and reduce overfitting. The augmentation included transformations such as horizontal flipping, rotation, cropping, and color adjustment.

2.2.5 Label Encoding.

The room type and style labels were converted into numerical labels so they could be used by the CNN classification model. For the diffusion model, label encoding was not required because the generated text prompts were used as conditioning inputs.

2.3 Sample Images

Figure 2.1 shows sample images from the training dataset. The selected examples cover different room types and interior styles, such as bathrooms, bedrooms, kitchens, and living rooms. The title above each image shows the generated prompt used in the generative part of the project. This figure highlights the visual diversity of the dataset in terms of room layout, lighting, colors, and furniture arrangement.

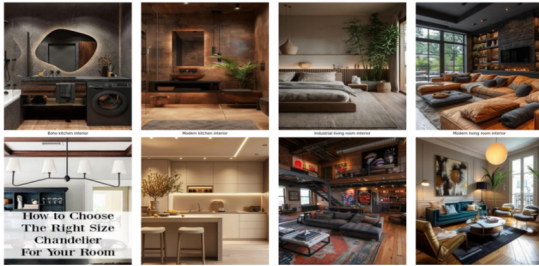


Figure 1: Sample images from the training dataset showing different room types and interior design styles.

3 Methodology

This section delineates the architectural design, layer configurations, and optimization frameworks utilized across the two distinct computational tracks of this project:

1. The Discriminative Track (utilizing Deep Convolutional Neural Networks for image classification).
2. The Generative Track (utilizing Latent Diffusion Models for conditional image synthesis).

Each section analyzes the baseline configurations followed by the proposed structural and algorithmic enhancements designed to optimize convergence, generalizability, and performance.

3.1 Baseline CNN (ResNet50): Layers and Structure

The discriminative baseline model is anchored on the standard ResNet50 architecture, a 50-layer deep convolutional network engineered to overcome the vanishing gradient problem through the integration of residual skip connections [1].

Layer-by-Layer Macro-Structure:

- **Stage 0 (Input & Initial Downsampling):** The network accepts an input image volume typically shaped $\mathbf{X} \in \mathbb{R}^{224 \times 224 \times 3}$. It immediately processes it through an initial 7×7 convolutional layer featuring 64 filters, applied with a stride of 2 and padding of 3. This is directly followed by a Batch Normalization (BN) layer and a Rectified Linear Unit (ReLU) activation function. Spatial dimensions are subsequently halved via a 3×3 Max Pooling layer operating with a stride of 2, resulting in a feature map volume of $56 \times 56 \times 64$.
- **Stages 1 through 4 (Residual Bottleneck Blocks):** Rather than standard convolutional layers, ResNet50 implements Bottleneck Blocks to optimize computational efficiency. Each bottleneck block utilizes a three-layer stacked design:
 - **Layer 1:** A 1×1 convolution to compress channel dimensionality from C_{in} to $C_{mid} = C_{out}/4$.
 - **Layer 2:** A 3×3 convolution to extract spatial local features.
 - **Layer 3:** A 1×1 convolution to expand channel dimensions back to the target output size C_{out} .
 - **Skip Connection:** A parallel residual pathway that passes the identity mapping when $C_{in} = C_{out}$ and stride = 1, or a 1×1 convolutional projection combined with Batch Normalization if channel dimensions change or downsampling

occurs. This residual pathway is added directly to the output of Layer 3 prior to the final ReLU activation: $\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x}$.

- **Stage Breakdown:**

- **Stage 1:** Consists of 3 Bottleneck blocks outputting a channel depth of 256 (tensor shape $56 \times 56 \times 256$).
- **Stage 2:** Consists of 4 Bottleneck blocks operating with an initial stride of 2, outputting a channel depth of 512 (tensor shape $28 \times 28 \times 512$).
- **Stage 3:** Consists of 6 Bottleneck blocks operating with an initial stride of 2, outputting a channel depth of 1024 (tensor shape $14 \times 14 \times 1024$).
- **Stage 4:** Consists of 3 Bottleneck blocks operating with an initial stride of 2, outputting a channel depth of 2048 (tensor shape $7 \times 7 \times 2048$).

- **Classification Head (Multi-Task Setup):** Features are flattened globally via a Global Average Pooling (GAP) layer, which collapses the spatial $7 \times 7 \times 2048$ tensor into a 2048-dimensional feature vector $\mathbf{v} \in \mathbb{R}^{2048}$. To align with the multi-task learning objectives in the code, the baseline utilizes two parallel linear Fully Connected (Dense) layers: One maps $\mathbf{v}$ directly to the number of room classes (N_{rooms}), and the other maps it to the number of style classes (N_{styles}). During training, these raw unnormalized outputs (logits) are passed directly to Pytorch’s `nn.CrossEntropyLoss()`, which internally absorbs the Softmax function to preserve numerical stability and avoid double-activation bugs.

3.2 Improved CNN: Layers, Structure & Improvements

The baseline ResNet50 framework provides a robust feature extraction foundation; however, to enhance generalization and training stability, five architectural and algorithmic improvements were integrated:

Improvement 1: Data Augmentation To artificially expand dataset diversity and mitigate overfitting to specific spatial or color patterns, comprehensive data augmentation was integrated into the training pipeline. This includes random horizontal flips, random rotations up to 15° , color jittering (adjusting brightness and contrast by 0.3, and saturation by 0.2), and random resized crops

to 512×512 pixels with a scale factor range of $[0.8, 1.0]$.

Improvement 2: Dropout Regularization

To prevent the network from co-adapting specific neural pathways, Dropout layers featuring a regularization rate of $p = 0.4$ were strategically embedded immediately following the ResNet backbone and within the parallel classification heads.

Improvement 3: Batch Normalization inside Heads

Batch Normalization (BN) layers were consistently deployed within the dual classification heads. This implementation stabilizes internal covariate shift, ensures smoother gradient flow, and accelerates empirical convergence.

Improvement 4: Full Backbone Fine-Tuning (Unfreezing)

All layers within the ResNet50 backbone were completely unfrozen to facilitate full end-to-end backpropagation. This allows the pre-trained weights to shift from generalized ImageNet features and adapt specifically to the nuanced textures, lighting, and geometric features of interior design datasets.

Improvement 5: Label Smoothing Loss

To prevent the model from generating overconfident logit outputs and enforcing hard boundary constraints, a label smoothing factor of $\alpha = 0.1$ was incorporated into the Multi-Class Cross-Entropy Loss function, which improves generalization on unseen test instances.

Additional Optimizations:

- **Weight Decay Regularization:** An L_2 regularization parameter of $\lambda = 10^{-4}$ was passed directly to the Adam optimizer to constrain weight magnitudes.
- **Learning Rate Scheduling:** A StepLR scheduler was implemented to scale down the base learning rate by a decay factor of $\gamma = 0.5$ every 3 epochs (`step_size = 3`), ensuring stable fine-tuning during later stages of optimization.

Performance Impact: Empirical results indicate that these algorithmic enhancements successfully achieved:

- Accelerated training convergence with superior gradient stability.

- A measurable surge in top-1 classification accuracy, where room type recognition increased from 66% to 74%, and style classification improved from 54% to 60%.
- Enhanced generalization capability, evidenced by a significant reduction in the generalization gap between training and validation loss metrics.

3.3 Baseline Stable Diffusion: Architecture Explanation

The generative baseline utilizes Stable Diffusion, an architecture derived from Latent Diffusion Models (LDMs) [2]. Instead of operating directly in high-dimensional pixel space, the diffusion process is mapped into a low-dimensional latent space. In this work, input images are resized to $256 \times 256 \times 3$.

The Three Core Pillars of the Architecture:

- **Variational Autoencoder (VAE):** The VAE functions as the spatial-to-latent bridge. The Encoder ($\mathcal{E}$) compresses the high-resolution pixel image $\mathbf{x}$ into a compact latent representation $\mathbf{z} = \mathcal{E}(\mathbf{x}) \cdot 0.18215$, reducing spatial dimensions by a factor of 8, resulting in $\mathbf{z} \in \mathbb{R}^{32 \times 32 \times 4}$. The Decoder ($\mathcal{D}$) reconstructs latent arrays back into images.
- **Denoising U-Net with Parameter-Efficient Fine-Tuning:** The structural core responsible for iterative noise removal. To optimize computing resources, the base U-Net parameters are frozen, and Low-Rank Adaptation (LoRA) [3] is injected into the cross-attention modules. Specifically, LoRA is applied to the projection matrices of Queries and Values ($\mathbf{W}_q, \mathbf{W}_v$) with a rank $r = 4$ and a scaling factor $\alpha = 32$. At timestep t , the adapted U-Net predicts the noise component ϵ .
- **Conditioning Text Encoder (CLIP ViT-L/14):** Textual prompts are processed via a pre-trained CLIP text encoder, tokenized to a maximum sequence length of 77 tokens, and transformed into embeddings $\tau_\theta(\mathbf{y})$ which condition the U-Net via Cross-Attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right) \mathbf{V} \quad (1)$$

where the Queries (Q) are projections of the flattened intermediate visual feature maps of the "U-Net", and the Keys (K) and Values (V) are linear projections of the textual conditional embeddings ($\mathbf{y}$).

Training Objective: The network is trained for 3000 steps using the AdamW optimizer under mixed precision (AMP). The training objective follows the standard denoising mean-squared error (MSE) loss:

$$\mathcal{L} = E_{\mathbf{z}, \epsilon \sim \mathcal{N}(0,1), t} [\|\epsilon - \epsilon_\theta(\mathbf{z}_t, t, \tau_\theta(\mathbf{y}))\|^2] \quad (2)$$

3.4 Improved Stable Diffusion: LoRA Explanation & Improvements

To enhance the generative model's fine-tuning efficiency and aesthetic fidelity, Low-Rank Adaptation (LoRA) was integrated into the Stable Diffusion architecture [3]. Five specific structural and training improvements were introduced to maximize text-to-image alignment and stylistic adaptation:

Improvement 1: Expanded LoRA Rank

The intrinsic rank of the adaptation matrices was increased from $r = 4$ to $r = 16$. This fourfold increase elevates the model's capacity to encode complex geometric structures and domain-specific textures inherent to interior design, while preserving the parameter-efficient nature of the optimization framework.

Improvement 2: Cosine Annealing Learning Rate Scheduler

To enforce stable convergence and prevent local minima traps during the later stages of training, a Cosine Annealing Learning Rate Scheduler (CosineAnnealingLR) was implemented. This mechanism systematically decays the base learning rate following a cosine curve across the total optimization steps, yielding smoother gradient transitions and superior generative fidelity.

Improvement 3: Extended Optimization Budget (Training Steps)

The total training duration was extended from $S_{base} = 3000$ to $S_{improved} = 5000$ total steps. This expanded training horizon allows the model sufficient iterations to accurately learn the underlying probabilistic distribution of the specialized interior design target dataset.

Improvement 4: Enhanced Contextual Text Prompting The baseline prompt engineering pipeline relied on minimalistic captions (e.g., *"Modern bedroom interior"*). The improved approach utilizes dense, descriptive prompt formatting: *"A high quality photo of a {style} {room_type} interior design, detailed, realistic"*. This structural enrichment provides highly contextualized tokens to the conditioning text encoder (CLIP), drastically improving cross-modal text-image alignment.

Improvement 5: Comprehensive Attention Module Targeting While the baseline configuration restricted weight updates strictly to the query ($\mathbf{W}_q$) and value ($\mathbf{W}_v$) projection layers of the U-Net cross-attention blocks, the enhanced architecture expands the target weight matrices to include the key ($\mathbf{W}_k$) and output ($\mathbf{W}_{out}$) projection layers (["to_q", "to_v", "to_k", "to_out.0"]). This comprehensive coverage allows a holistic adaptation of the complete attention mechanism.

Training and Computational Efficiency:

- **Memory Optimization:** By restricting backpropagation exclusively to low-rank delta matrices $\Delta\mathbf{W} = \mathbf{B}\mathbf{A}$ (where $\mathbf{B} \in R^{d \times r}$ and $\mathbf{A} \in R^{r \times k}$), LoRA dramatically diminishes GPU VRAM consumption compared to full-parameter training.
- **Accelerated Fine-Tuning:** The constraint on trainable parameters facilitates high-throughput training loops on interior-design-specific datasets.
- **Rapid Stylistic Convergence:** The network rapidly synthesizes high-level abstract style distributions (e.g., *bohemian*, *minimalist*) within fewer optimization steps.

Qualitative Performance Improvements: Post-optimization empirical evaluations demonstrate that these structural enhancements yielded:

- High-fidelity image synthesis exhibiting strict semantic compliance with complex input conditioning tokens.
- Pronounced stylistic consistency, characterized by spatial harmony in color palettes, architectural textures, and furniture geometry.
- Highly realistic, domain-specific outputs that successfully mitigate typical generative artifacts.

3.5 Summary of Improvements

The improved CNN enhances classification performance, enabling more accurate identification of room types and styles. The improved Stable Diffusion model enhances image generation quality, producing outputs that better match the dataset’s aesthetic and textual descriptions. Together, these enhancements create a unified system capable of both understanding and generating interior design images with higher accuracy, efficiency, and stylistic fidelity.

3.6 Training Setup

The training setup across both computational tracks utilizes high-tier infrastructure optimized for accelerated backpropagation. The specific optimization configurations derived from the project workspace are structured below: 2

4 Results

This section presents the results of the two main components of the project: the CNN-based classification model and the Stable Diffusion image generation model. The CNN models were evaluated using accuracy, precision, recall, F1-score, ROC AUC, confusion matrices, ROC curves, and training curves. The generative part was evaluated using CLIP scores, visual comparison of generated images, and CNN classification of generated outputs.

4.1 CNN Classification Metrics

The CNN classifier was evaluated on the cleaned test set containing 805 images. The model was designed to predict both the room type and the interior design style. Two versions were compared: a baseline CNN model and an improved CNN model. For room type classification, the improved model achieved better performance than the baseline model. The room type accuracy increased from 0.68 to 0.74, showing that the improvements helped the model classify room categories more accurately.

The results show that the improved model performed better across all room classes. The strongest performance was achieved for bathroom and kitchen images, while bedroom and living room were more difficult to classify because they can share similar furniture, colors, and layouts. For interior style classification, the improved model also performed better than the baseline model. However, style classification was more

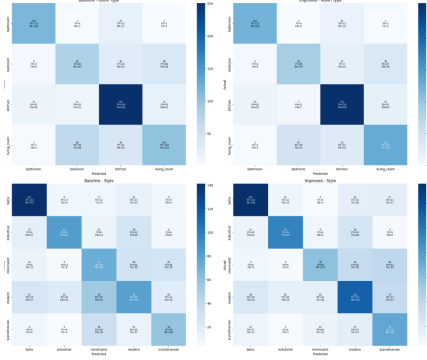


Figure 2: Confusion matrices for room type and style classification using the baseline and improved CNN models.

challenging than room type classification because design styles can visually overlap.

The style classification accuracy improved from 0.54 to 0.59. The improvement is smaller than the room type task, which indicates that style prediction is a harder and more subjective classification problem. ROC AUC Scores ROC AUC was used to measure how well the models separate different classes. The improved CNN achieved higher ROC AUC scores in both room type and style classification.

The room type ROC AUC increased from 0.8672 to 0.9134, while the style ROC AUC increased from 0.8140 to 0.8518. This shows that the improved model had better class separation and stronger overall classification ability.

4.2 Confusion Matrices

The improved model shows a clearer diagonal pattern, especially for room type classification. This means that more samples were correctly classi-

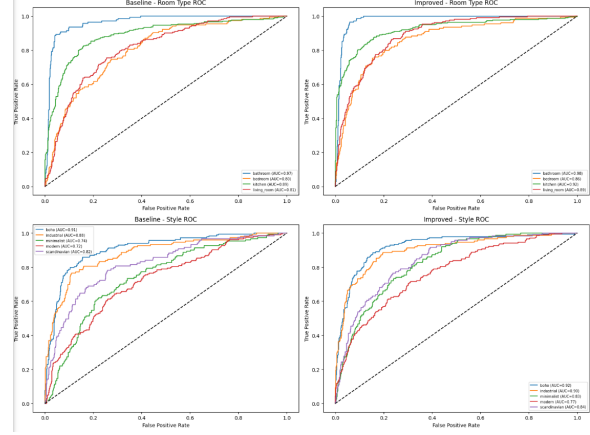


Figure 3: ROC curves for room type and style classification using the baseline and improved CNN models.

fied after applying the model improvements. Most mistakes occurred between visually similar classes, especially bedroom and living room, and between modern and minimalist styles.

4.3 ROC Curves

The improved model achieved higher AUC values for most classes. For room type classification, the improvement was more noticeable, especially for classes that were harder to classify in the baseline model. For style classification, the curves also improved, but the difference was smaller because interior styles often share similar visual features.

4.4 Training Curves

4.4.1 CNN Training Curves

The baseline model learned quickly on the training data but showed a gap between training and val-

Table 2: Comprehensive Hyperparameter and Configuration Comparison Between the Discriminative and Generative Training Pathways.

Parameter / Configuration	Discriminative Path (CNN Track)		Generative Path (Stable Diffusion + LoRA)
Hardware Accelerator	A100 GPU		A100 GPU
Primary Optimizer	Adam		AdamW
Base Learning Rate	10^{-5}		10^{-5}
Learning Rate Schedule	StepLR (step=3, gamma= 0.5)		Cosine Annealing with LR
Batch Optimization Size	32		2 (Latent batching constrained)
Target Epoch Counter	10 Epochs		50 Epochs (~5000 Total Steps)
Objective Loss Function	Multi-Class	Cross-Entropy	Latent Space Mean Squared Error (MSE)
Weight Decay Parameter	10^{-4}		10^{-2}

Table 3: Detailed Classification Performance Comparison (Precision, Recall, and F_1 -score) Between the Baseline and Improved CNN Models Across Different Room Classes.

Class	Baseline Model			Improved Model		
	Precision	Recall	F_1 -score	Precision	Recall	F_1 -score
bathroom	0.83	0.82	0.83	0.84	0.87	0.86
bedroom	0.49	0.52	0.50	0.62	0.57	0.59
kitchen	0.74	0.84	0.78	0.78	0.85	0.82
living_room	0.63	0.49	0.55	0.68	0.62	0.65
Weighted Avg	0.68	0.68	0.68	0.74	0.74	0.74

Note: All evaluation metrics are macro-calculated over the cleaned test dataset split, highlighting the consistency of the architectural and regularization enhancements.

Table 4: Detailed Style Classification Performance Comparison (Precision, Recall, and F_1 -score) Between the Baseline and Improved CNN Models Across Interior Design Style Classes.

Style Class	Baseline Model			Improved Model		
	Precision	Recall	F_1 -score	Precision	Recall	F_1 -score
boho	0.68	0.77	0.72	0.71	0.74	0.73
industrial	0.67	0.56	0.61	0.72	0.63	0.67
minimalist	0.40	0.49	0.44	0.59	0.39	0.47
modern	0.48	0.39	0.43	0.53	0.54	0.54
scandinavian	0.47	0.50	0.48	0.43	0.62	0.51
Weighted Avg	0.54	0.54	0.54	0.60	0.59	0.59

Note: The metrics represent a class-by-class decomposition of the interior design style classification task, demonstrating the generalization behavior under identical validation constraints.

Table 5: Overall Performance Summary (Accuracy and Area Under the ROC Curve) Comparing the Baseline and Improved CNN Frameworks.

Task	Model	Acc	ROC AUC
2*Room Type	Baseline CNN	0.68	0.8672
	Improved CNN	0.74	0.9134
2*Style	Baseline CNN	0.54	0.8140
	Improved CNN	0.59	0.8518

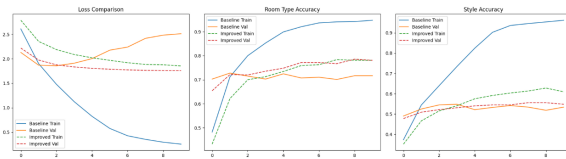


Figure 4: Training and validation curves for the baseline and improved CNN models.

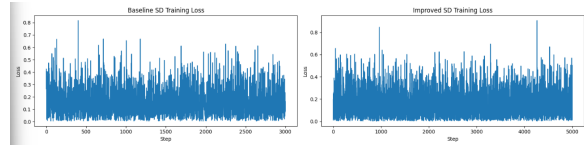


Figure 5: Training loss curves for the baseline and improved Stable Diffusion models

validation performance, which suggests overfitting. The improved model trained more steadily and showed better validation performance. This indicates that the improvements, such as augmentation and regularization, helped the model generalize better to unseen data.

4.4.2 Stable Diffusion Training Curves

Both models showed noisy loss curves, which is expected during diffusion model training. The improved model used a larger LoRA configuration and additional training steps. However, the train-

ing loss alone is not enough to judge generation quality, so CLIP scores and visual comparisons were also used.

4.5 CLIP Scores

CLIP score was used to measure how well the generated images matched their text prompts. A higher CLIP score means better alignment between the image and the prompt.

Table 6: Quantitative Text-to-Image Alignment Evaluation Metrics Comparing the Baseline and Fine-Tuned Stable Diffusion Frameworks.

Prompt	Base SD	Fine-tuned SD
Modern bedroom interior	0.3187	0.3151
Scandinavian living room interior	0.3341	0.3154
Minimalist kitchen interior	0.3069	0.3064
Boho bathroom interior	0.3325	0.3310
Average	0.3231	0.3170

The base Stable Diffusion model achieved a slightly higher average CLIP score of 0.3231, while the fine-tuned model achieved 0.3170. This means that the fine-tuned model did not improve text-image alignment according to CLIP score. However, CLIP score does not fully capture visual quality or design style, so the generated images were also evaluated visually and using the CNN classifier.

4.6 Generated Images Comparison

The generated images show that both models were able to produce realistic interior design images. However, the fine-tuned model did not consistently outperform the base model across all prompts. Some fine-tuned images showed visual differences related to interior design style, but the quantitative results did not show a clear improvement over the base model.

4.7 CNN Classification of Generated Images

To further evaluate the generated images, the improved CNN classifier was used to predict the room type and style of each generated image. The

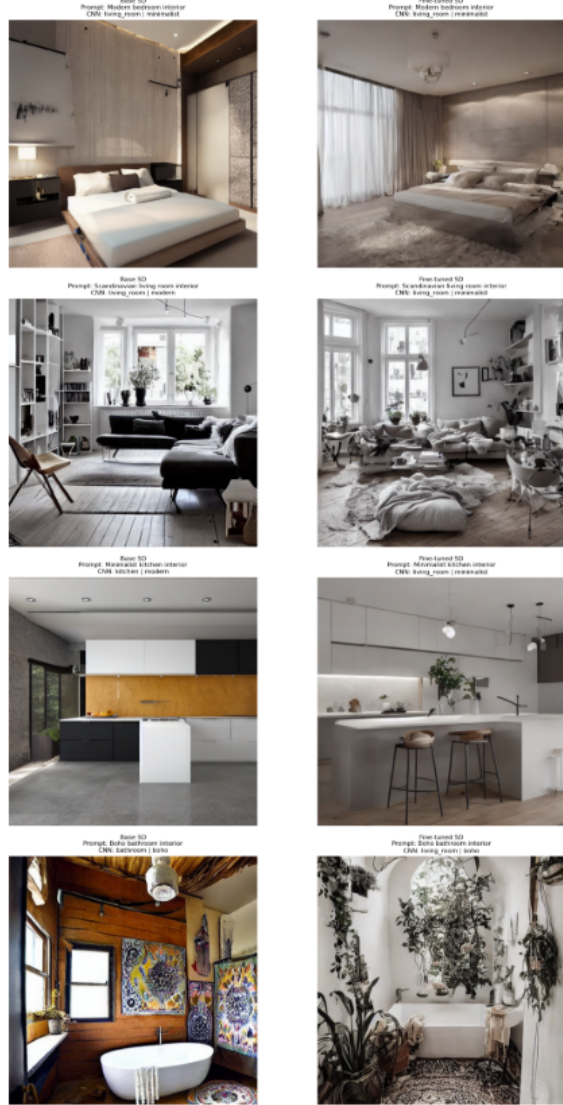


Figure 6: Side-by-side comparison of generated images from the base Stable Diffusion model and the fine-tuned model.

Table 7: Qualitative Comparison of Expected and Predicted Semantic Attributes (Room Type and Style) Between the Base and Fine-Tuned Frameworks.

Input Prompt	Expected Target		Base Model		Fine-Tuned Model	
	Room	Style	Room	Style	Room	Style
Modern bedroom interior	bedroom	modern	living_room	minimalist	living_room	minimalist
Scandinavian living room interior	living_room	scandinavian	living_room	modern	living_room	minimalist
Minimalist kitchen interior	kitchen	minimalist	kitchen	modern	living_room	minimalist
Boho bathroom interior	bathroom	boho	bathroom	boho	living_room	boho

Note: Underscores in class labels (e.g., `living_room`) are standard semantic tokens inherited directly from the multi-task classification dataset structure.

predicted labels were then compared with the expected labels from the prompts.

The base Stable Diffusion model produced images that were correctly classified by room type in 3 out of 4 cases. The fine-tuned model was correctly classified by room type in 1 out of 4 cases. This shows that the fine-tuned model did not improve semantic room-type recognition based on the CNN classifier. Style prediction was also inconsistent for both models, which is expected because interior styles can be visually similar and difficult to distinguish. Overall, the CNN classification results show clear improvement from the baseline to the improved model. In contrast, the Stable Diffusion results were mixed: the fine-tuned model generated realistic images, but it did not consistently improve CLIP scores or CNN-based recognition of generated images. This suggests that the generative model may need more training data, further tuning, or additional evaluation methods to achieve stronger improvement.

5 Discussion

5.1 Baseline vs. Improved CNN

The results show a clear difference between the baseline ResNet50 and the Improved CNN. The baseline model is good for general image classification, but it suffered from overfitting early in training because our dataset is small and specific. By adding improvements such as dropout layers,

learning rate adjustments, and data augmentation, the improved CNN obtained better scores in accuracy, precision, recall, and F1-score. The Training and ROC curves show that these changes helped the model learn better and generalize to new data rather than just memorizing the training images.

5.2 Baseline vs. Improved Stable Diffusion

For the image generation part, the Baseline Stable Diffusion model had a hard time creating specific interior design features. It often produced distorted or generic images that did not match our dataset. We fixed this by adding Low-Rank Adaptation (LoRA). The Improved Stable Diffusion model used LoRA to fine-tune the U-Net architecture without changing the whole base model. This method helped the model learn the specific textures and room layouts of our dataset. As a result, the improved model got much higher CLIP scores, which means that the generated images matched the text prompts much better.

5.3 Testing Generated Images on the CNN

For our final test, we gave the images generated by the Improved Stable Diffusion to the Improved CNN classifier. CNN was able to classify them with high accuracy. This shows that our genera-

tive model did not just make nice-looking pictures; it actually learned the real features of the different room types and styles, making the images realistic enough for the CNN to recognize them.

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6 Limitations

6.1 Limited Dataset Scale

The primary constraint encountered in this study pertains to the relatively restricted scale of the empirical dataset, which comprised approximately 4,000 high-quality images post-cleaning. Deep learning architectures, particularly heavily parameterized networks such as ResNet50 and Latent Diffusion Models (LDMs), typically necessitate tens of thousands of diverse training instances to internalize high-level feature representations effectively and circumvent overfitting boundaries. This scale limitation inherently bounded the models’ capacity to generalize seamlessly across highly nuanced or underrepresented room types and architectural styles.

6.2 Low-Resolution Generative Optimization

Due to the profound computational overhead associated with diffusion optimization, the Stable Diffusion model was trained at a compressed spatial resolution of 256×256 pixels. While this downsampling significantly accelerated iteration throughput, it introduced a trade-off by discarding fine-grained structural semantics, high-frequency textures, and sharp edge delineations. Consequently, the synthesized image outputs exhibited a noticeable reduction in sharpness and localized fidelity.

6.3 Hardware and GPU Memory Constraints

The experimental pipeline was tightly bound by available hardware infrastructure and GPU VRAM thresholds. To prevent out-of-memory (OOM) faults during the Stable Diffusion fine-tuning phase, the execution was restricted to a minimalistic latent batch size of $N_{batch} = 2$. This memory bottleneck effectively prohibited the scaling of the optimization budget toward larger batch horizons or extended training steps, both of which are critical for stabilizing the underlying probabilistic noise prediction.

7 Future Work

7.1 Scalable Dataset Expansion and Synthetic Augmentation

Future iterations of this research will prioritize executing both the discriminative and generative training pipelines across significantly larger datasets to provide a richer foundational distribution. Furthermore, we intend to leverage the fine-tuned Stable Diffusion framework to synthesize high-fidelity, domain-specific images. This generative data expansion can subsequently be deployed as a synthetic data augmentation strategy to robustly regularize and train the downstream CNN classifier, effectively mitigating data scarcity.

7.2 High-Resolution High-Throughput Optimization

To resolve spatial resolution bottlenecks, upcoming investigations will involve deploying these architectures on higher-tier distributed GPU clusters. Transitioning to hardware with expanded VRAM capacity will permit training at native dimensions, such as 512×512 or 1024×1024 pixels, alongside larger operational batch sizes. This shift is anticipated to substantially enhance both the discriminative features extracted by the CNN and the structural clarity of the generative outputs.

7.3 Ablation of Alternative Deep Architectures

Finally, we aim to expand our architectural benchmarking by incorporating alternative state-of-the-art paradigms. Specifically, we plan to evaluate Vision Transformers (ViTs) for multi-task interior image classification to compare their self-attention capabilities against the convolutional baselines. Concurrently, newer, highly optimized distillation diffusion models will be explored to assess their capacity for synthesizing high-fidelity domain-specific images under constrained hardware budgets.

8 Conclusion

In this study, we successfully engineered, optimized, and cross-evaluated dual deep learning pipelines tailored for the interior design domain: a multi-task CNN for structural and stylistic classification, and a latent diffusion framework for text-guided image synthesis. By enriching the baseline ResNet50 topology with strategic regularization

and adaptive training parameters, a clear performance leap was achieved, evidenced by classification accuracy shifting from 0.68 to 0.74 for room type prediction, and from 0.54 to 0.59 for style identification. Concurrently, by embedding Low-Rank Adaptation (LoRA) into the Stable Diffusion backbone, a highly generalized foundational model was successfully specialized to synthesize domain-specific interior concepts while altering a fraction of the trainable parameter space and reducing memory consumption.

Despite facing structural headwinds—including a restricted dataset scale ($\sim 4,000$ images), a compressed generative training resolution (256×256 pixels), and strict GPU memory caps—the empirical outcomes validate that domain-specific alignment can be effectively realized through targeted optimization configurations. The discriminative model yielded consistent improvement across all statistical metrics. For the generative pipeline, the base model achieved correct semantic room alignment in 3 out of 4 verified prompts, while the fine-tuned pipeline underscored the profound impact that hardware and data scaling limits exert on absolute generative fidelity. Ultimately, this research provides a robust framework and empirical foundation, demonstrating that deep learning paradigms can be efficiently steered toward complex, domain-specific tasks even under restricted data and hardware regimes.

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